

COVID-19 Exercise: Detecting Exponential Growth

SISMID 2026 • Day 2, 3:30 • COVID-19 exercise

In this exercise you use digital traces to detect outbreaks of COVID-19 in Washington State. There are many ways to define an outbreak; here you use a simple, useful one: **sustained exponential growth** of the signal. Download `covid_traces_WA.csv`. The first non-date column, `new_cases`, is the ground truth (positive tests). The remaining columns are digital traces: `upToDate` (clinician search), `cdc_ili`, `Twitter_RelatedTweets`, `google_fever`, `Kinsa` (smart-thermometer fever), and `Cuebiq_Mobility`. The data is pre-smoothed, so case counts are fractional.

You may work in **Python or R**, and either drive a coding agent (Codex, Claude Code, or Antigravity CLI) with a prompt per part, or fill in the pre-filled notebook. Data + solution: <https://github.com/MIGHTE-lab/SISMID25>.

- (a) **Look first.** In a 6-panel plot, show the case counts and the five other traces over time. By eye, where do outbreaks start? Do any traces show “outbreak” behavior, and are they leading or lagging relative to cases?
- (b) **A growth factor.** Slide an 11-day window over `new_cases`. In each window, fit a linear regression of the last 10 days on the previous 10 days, **with no intercept**. Collect the slope α_i for each day (the first ten or so will be 0, which is fine). α_i is the multiplicative growth factor: $\alpha_i > 1$ means the signal is growing.
- (c) **Read it.** Make a 3-panel plot: (i) the case curve, (ii) α over time with a reference line at 1, and (iii) a 0/1 flag for $\alpha_i > 1$. Interpret α . (Note $\alpha \approx e^r$, so the doubling time is $\ln 2 / \ln \alpha$.)
- (d) **Detect outbreaks.** When $\alpha_i > 1$ for 10 consecutive days (sustained growth), mark an outbreak as active starting on the 10th day, but record only the *first* such day. Once there are 10 consecutive days without growth ($\alpha_i < 1$), the outbreak is over; a later run of growth starts a new one. Implement this and apply it to the case series.
- (e) **Mark the starts.** Redo the 3-panel plot from (c) with markers at the detected outbreak starts. Do they match your intuition from (a)?
- (f) **Multi-trace early warning.** Repeat (b)-(e) for each of the other five traces. Redraw the 6-panel plot with each trace’s outbreak starts marked. For each trace, report how many days before or after the case-count outbreak its own outbreak fires. If enough traces flag growth ahead of cases, you can predict an imminent outbreak.

Discuss: which traces are the most reliable leading indicators, and what would make you trust (or distrust) a “taking off” alarm built from them?